

Data Engineer Interview Question Bank

Databricks | Spark / PySpark | Delta Lake | CDC | Power BI | SQL | Data Modeling

Purpose: A focused revision sheet based on the interview topics and questions you shared.

Total questions: 39

Most Important Topics They Tested

- Spark / PySpark
- Databricks / Delta Lake
- Optimization
- CDC
- Power BI / RLS
- SQL
- Data Modeling
- OLTP / OLAP

Project / Architecture

1. Can you explain one of your projects' complete architecture, including the end-to-end flow of data from the source system to where it is used?
2. How were you able to capture CDC data?
3. How did your CDC process handle inserts, updates, and deletes?
4. What is the role of the tracker table and offset in your CDC process?

Databricks / Spark Optimization

5. What optimization techniques do you use within your projects?
6. Can you divide your optimization techniques into table-level optimization and execution-level optimization?
7. What joining strategies do you use in Spark?
8. What is AQE in Spark?
9. What is the structure of a Delta table?
10. Suppose a CDC table was initially fast to query but became slow after a few months. Why would this happen and how would you resolve it?
11. Suppose a Spark job is taking a long time to execute. What steps would you follow to optimize it?
12. What is a Liquid Cluster in Databricks?

Databricks Compute / Cluster

13. You have a job that runs every five minutes and takes one minute to execute. Which Databricks cluster would you choose?
14. What would you do if the cluster takes six minutes to spin up while the job needs to run every five minutes?

Power BI / RLS

15. You mentioned that you worked with RLS. Did you implement it in Databricks or Power BI?
16. Did you use the USERPRINCIPALNAME() function for RLS?
17. Did you create one role using DAX and one mapping table?
18. Can you walk me through the exact use case of RLS in that particular project or sprint?

APIs / Streaming

- 19. You mentioned that you worked with APIs. Did you use an extractor into Databricks, a data lake, or something else?
- 20. Did you work with streaming?
- 21. Did you use Auto Loader?

Python Coding

- 22. Find the leaders in an array.
Example: [16, 17, 4, 3, 5, 2] -> [17, 5, 2].

SQL

- 23. Given customers and orders tables, find the customers who made the second-highest order, returning their first name and last name.
- 24. How would you handle the case where multiple customers have the same second-highest order?

Schema / Delta

- 25. How do you handle schema evolution within Spark, Databricks tables, or the jobs you create?
- 26. Why did you use mergeSchema = false?
- 27. What is the difference between mergeSchema and Delta MERGE INTO?
- 28. What are the different options available in the MERGE INTO command?

Spark Concepts

- 29. What is the Catalyst Optimizer in Spark?
- 30. What is lazy evaluation in Spark?
- 31. What are transformations and actions in Spark?
- 32. How does Spark optimize a query before execution?

Data Schema / Modeling

- 33. Have you worked with data schemas?
- 34. Have you worked with data modeling?

35. What is a star schema?

36. What is the difference between a fact table and a dimension table?

37. Should a particular property/attribute go into a dimension or fact table?

Database Concepts

38. Do you work with both OLTP and OLAP systems?

39. What is the difference between OLTP and OLAP?

Revision Focus

Be ready to connect theory to your own project: explain the architecture first, then justify design choices such as CDC, tracker offsets, Delta MERGE, cluster selection, partitioning/clustering, Spark joins, schema evolution, and production optimization.